實驗設計：

本實驗採用自行實作的 ResNet-18 型 CNN，從隨機初始化開始訓練加入 Dropout(p = 0.3)與全連接層，輸出六個類別的分類分數。

資料前處理方面，訓練影像使用 RandomResizedCrop，隨機裁切原圖面積的 80%～100%，長寬比設定為 0.9～1.1，並縮放至 128 * 128。另外，以 0.5 的機率進行水平翻轉，並透過 ColorJitter 調整亮度、對比度與飽和度，三者參數皆設為 0.2，以增加訓練資料的變化。驗證與測試影像則只縮放至 128 * 128。

訓練分為 supervised learning 與 self-training 兩個階段。第一階段僅使用有標籤的訓練資料，batch size 設為 64，損失函數使用 CrossEntropyLoss，optimizer 使用 Adam，learning rate 為 1e-3，weight decay 為 1e-4。

第二階段延續第一階段訓練後的模型權重，利用模型預測未標記影像。將輸出經過 softmax 轉為類別機率，選取最高機率達到 0.9 的預測作為 pseudo-label，並將這些影像加入訓練資料。Self-training 每隔 10 epochs 重新預測剩餘未標記影像，將新接受的資料與原始有標籤資料、先前累積的 pseudo-labeled data 合併後繼續訓練；已接受的 pseudo-label 保持不變。此階段將 learning rate 降為 1e-4，其餘主要設定維持相同，讓模型以較小的更新幅度利用新增資料繼續學習。

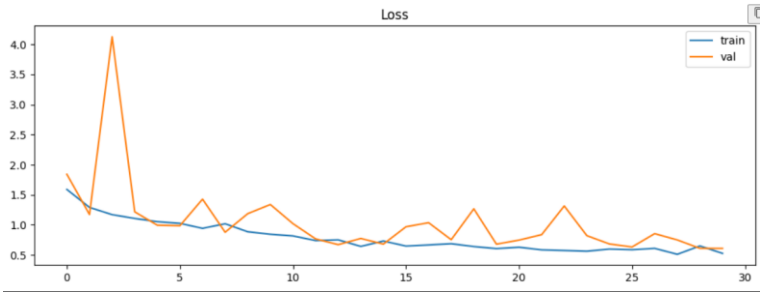
Activation function 結果比較：

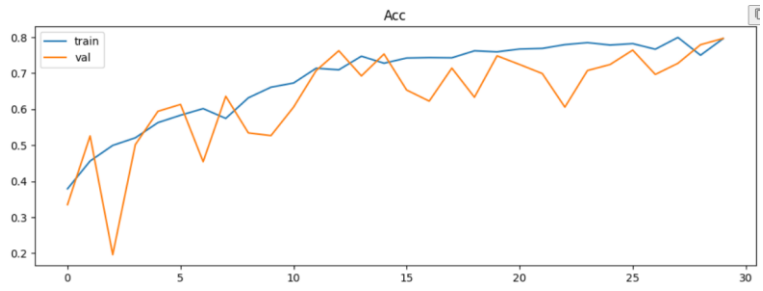
Kaggle 分數最高的是 GeLU，接著是 LeakyReLU，最後是 ReLU；在 supervised model 與 self train model 表現上，validation loss 與 validation accuracy 表現最好的也是 GeLU，同時也標記出最多 pseudo label，表現次好的則是 ReLU。

ReLU：Kaggle score = 0.84466

```
==================== Epoch 20/30 ====================
Train Acc: 0.759444 | Train Loss: 0.605549
  Val Acc: 0.748333 |   Val Loss: 0.679769 | LR: 0.001000

==================== Epoch 30/30 ====================
Train Acc: 0.796111 | Train Loss: 0.526685
  Val Acc: 0.796667 |   Val Loss: 0.609011 | LR: 0.001000
```






```
==================== Epoch 40/50 ====================
Train Acc: 0.949430 | Train Loss: 0.145674
  Val Acc: 0.866667 |   Val Loss: 0.456180 | LR: 0.000100
Dataset Size: 4647 (original: 1800, pseudo: 2847)

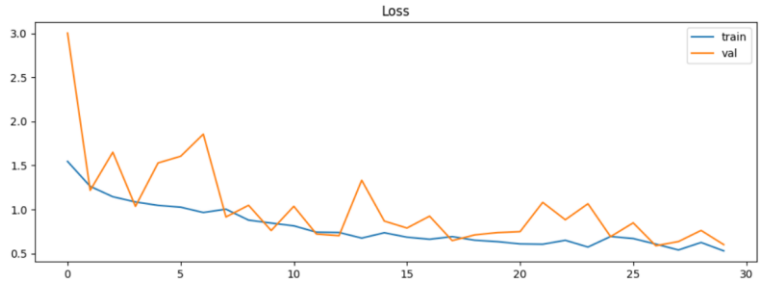
Labeled 135 images this round. Remaining unlabeled: 618

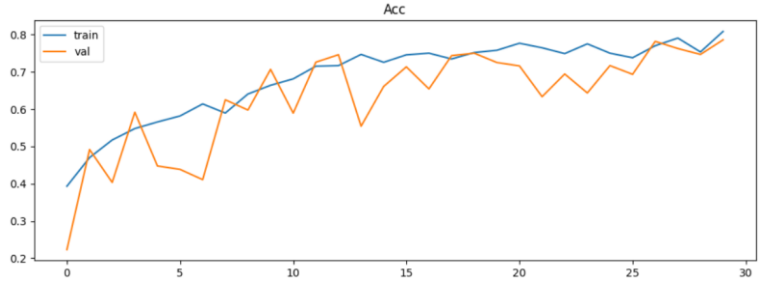
==================== Epoch 50/50 ====================
Train Acc: 0.951694 | Train Loss: 0.132426
  Val Acc: 0.845000 |   Val Loss: 0.510520 | LR: 0.000100
Dataset Size: 4782 (original: 1800, pseudo: 2982)
Best validation accuracy: 0.8666666666666667
```

LeakyReLU：Kaggle score =0.84600

```
==================== Epoch 20/30 ====================
Train Acc: 0.757778 | Train Loss: 0.635776
  Val Acc: 0.725000 |   Val Loss: 0.738387 | LR: 0.001000

==================== Epoch 30/30 ====================
Train Acc: 0.808333 | Train Loss: 0.531246
  Val Acc: 0.785833 |   Val Loss: 0.602193 | LR: 0.001000
```





```
==================== Epoch 40/50 ====================
Train Acc: 0.928716 | Train Loss: 0.202744
  Val Acc: 0.847500 |   Val Loss: 0.461466 | LR: 0.000100
Dataset Size: 2960 (original: 1800, pseudo: 1160)

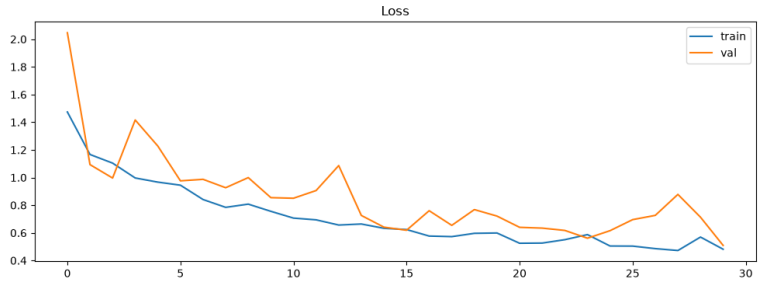
Labeled 180 images this round. Remaining unlabeled: 629

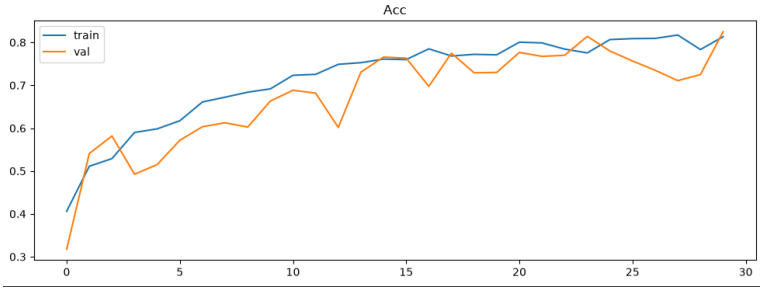
==================== Epoch 50/50 ====================
Train Acc: 0.935669 | Train Loss: 0.180964
  Val Acc: 0.847500 |   Val Loss: 0.470518 | LR: 0.000100
Dataset Size: 3140 (original: 1800, pseudo: 1340)
Best validation accuracy: 0.865
```

GeLU：Kaggle score = 0.85600

```
==================== Epoch 20/30 ====================
Train Acc: 0.771111 | Train Loss: 0.598934
  Val Acc: 0.730000 |   Val Loss: 0.721101 | LR: 0.001000

==================== Epoch 30/30 ====================
Train Acc: 0.813333 | Train Loss: 0.481559
  Val Acc: 0.825000 |   Val Loss: 0.509252 | LR: 0.001000
```






```
==================== Epoch 40/50 ====================
Train Acc: 0.955164 | Train Loss: 0.131442
  Val Acc: 0.866667 |   Val Loss: 0.467674 | LR: 0.000100
Dataset Size: 4773 (original: 1800, pseudo: 2973)

Labeled 156 images this round. Remaining unlabeled: 471

==================== Epoch 50/50 ====================
Train Acc: 0.960844 | Train Loss: 0.112416
  Val Acc: 0.871667 |   Val Loss: 0.472012 | LR: 0.000100
Dataset Size: 4929 (original: 1800, pseudo: 3129)
Best validation accuracy: 0.88
```